

Digital System Design

Dr. Nazanin Ghasemian

Fall 2026 · Lecture 1



Lecture, Lab, and Office Hours



Lecture Time

Tuesdays and Thursdays, 4:00 – 5:15 PM

Online via Zoom



Lab Time

Tuesdays and Thursdays, 5:30 – 6:45 PM

Online via Zoom



Office Hours

By appointment

Structure and Grading



Assignment/Quizzes 30% + Lab 30% + Final Project/Presentation 40% = 100%

References — Textbooks

1

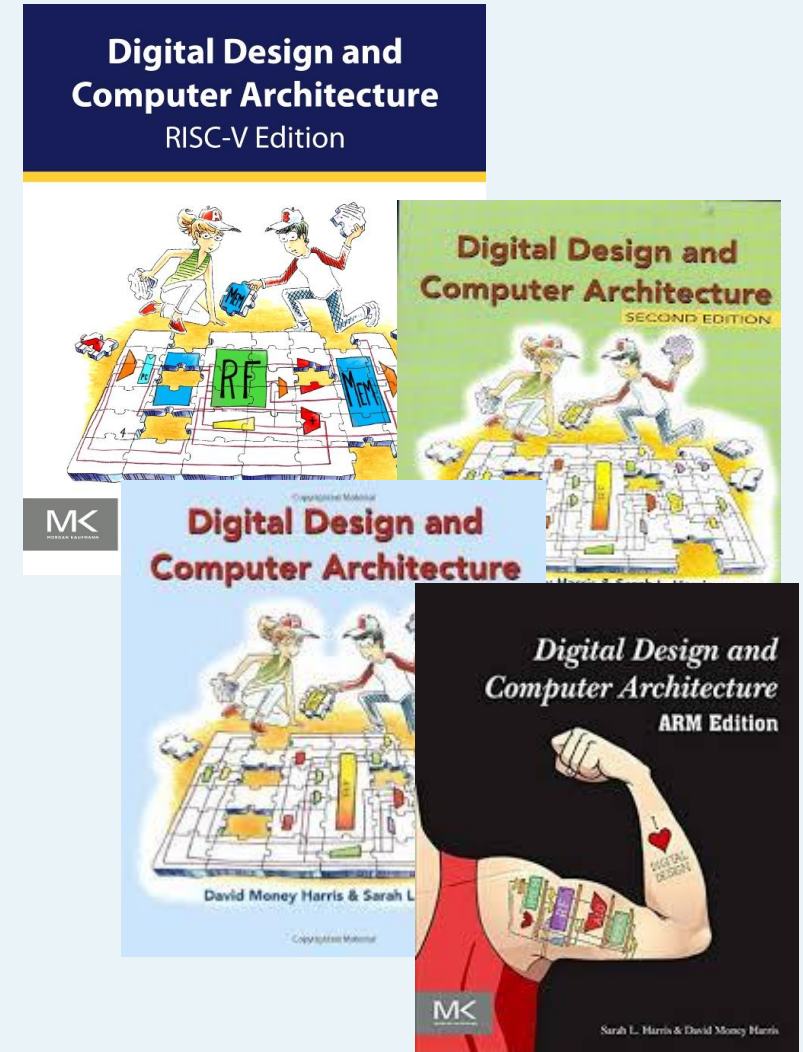
Sarah L. Harris, David Harris. Digital Design and Computer Architecture, RISC-V Edition, Morgan Kaufmann, 2022.

2

Sutherland, Stuart, Simon Davidmann, and Peter Flake. SystemVerilog for Design. Springer, 2004.

3

Spear, Chris. SystemVerilog for Verification. Springer, 2006.



References — Online



[Doulos SystemVerilog Tutorials](#)



[ChipVerify SystemVerilog Tutorials](#)



[ASIC World SystemVerilog Tutorials](#)

Introduction

Everything in our digital world — computers, phones, cars, even automatic doors — is built from digital circuits.



Functions as Truth Tables

- Any digital system starts with a function.
- For example: a light should turn ON if the motion sensor detects someone and it's nighttime.
- In logical terms: $\text{Light} = \text{Motion AND Night}$.

TRUTH TABLE

Motion	Night	Light
0	0	0
0	1	0
1	0	0
1	1	1

Combinational vs. Sequential

Combinational

Output depends only on the current input.

EXAMPLES

Adders · Multiplexers · Comparators

Sequential

Output depends on the current input plus previous state (memory).

EXAMPLES

Flip-Flops · Counters · Registers

Sequential circuits let us design systems with timing and memory, like traffic lights or digital clocks.

Scaling Up (Ex. Processors)



Start with logic gates → build adders → group them to form an Arithmetic Logic Unit (ALU).

Add a control unit and registers → you have a CPU.

So, at its heart, a processor is just a very large collection of digital circuits.

Processor Architecture Basics

Core components:

ALU

Arithmetic Logic Unit — does math/logic.

CU

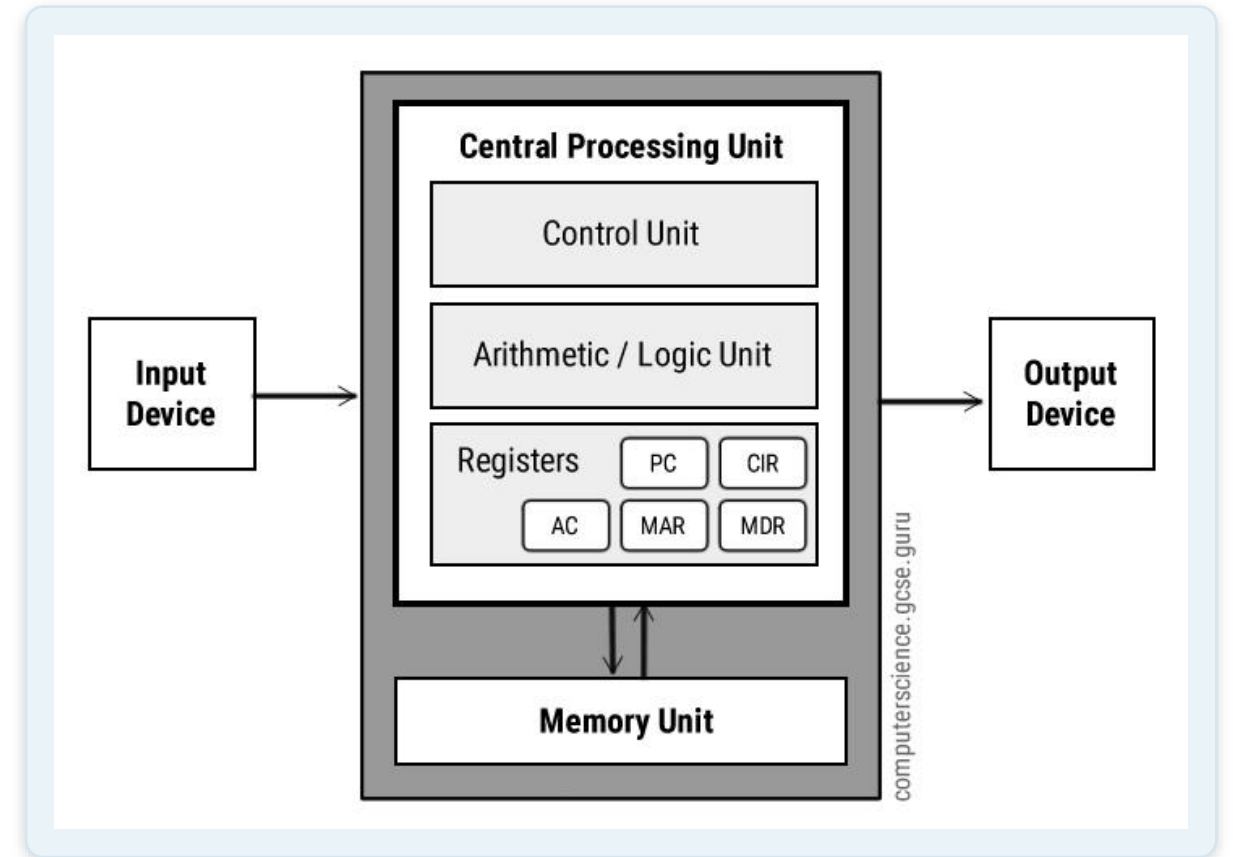
Control Unit — decides which operation to perform.

REG

Registers — small, fast memory inside the CPU.

I/O

Memory + I/O — store and move data.



ASIC (Application-Specific Integrated Circuit)

A chip designed for one job — e.g., a WiFi modem, GPU, or iPhone FaceID.

PROS

- Fast
- Power efficient
- Optimized for its task

CONS

- Expensive to design
- Fixed forever once manufactured

FPGA (Field-Programmable Gate Array)

A reconfigurable chip — we can “program” it with our own digital circuits.

FLEXIBLE

Change the design anytime, and test quickly.

USED IN

Prototyping

Research

Aerospace

High-Speed Trading (Finance)



Specification

Define what the system should do.

EXAMPLE

Automatic door should open when motion is detected, close after 5 seconds, and stop if an obstacle is detected.

Design (HDL)

Write code in a Hardware Description Language (HDL) such as Verilog or VHDL.

Example snippet (pseudo-Verilog):



```
if (motion && !obstacle)
    door = OPEN;
else
    door = CLOSE;
```

Simulation & Synthesis

3 Simulation

Run the HDL in a simulator to test the logic with virtual inputs.

Check: does the door open and close correctly in different cases?

4 Synthesis

Translate the HDL code into actual FPGA logic blocks (LUTs, flip-flops).

Implementation & Testing

5

Implementation (Place & Route)

Map logic into the FPGA's physical structure — where each flip-flop and connection goes.

6

Programming / Testing

- Upload the design onto FPGA hardware.
- Connect real sensors and actuators.
- Observe the door actually opening and closing.